

Research Agenda and Keystone Positivity Report

Executive Summary

The request describes the subject as unspecified, but the attached research brief supplies a concrete, unusually well-defined default target: an **open all-orders positivity problem for Jacobi/partial-wave coefficients in a one-parameter family of quantum-gravity S-matrices**. The target is to prove, for every level $n \geq 3$, every trajectory index $3 \leq j \leq n - 1$, every $\lambda > 0$, and all physical dimensions $D \leq \hat{T}(\lambda)$, that the signed Jacobi coefficient called knife_j is nonnegative. The brief reports exact low-depth proofs through depth $d = j - 1 = 6$, an exact fixed- j scaling limit, a certified dimension-monotonicity component, extensive falsification of several tempting structural conjectures, and a remaining need for a mechanism **uniform in j** . [filecite turn0file0](#)

I recommend making that **keystone positivity problem the immediate priority**. It has unusually high research leverage: a theorem would replace a sequence of depth-by-depth certificate computations with a structural all-orders result; the existing exact engine makes new conjectures cheaply falsifiable; and the already-derived scaling law tells us what the all-orders mechanism should reduce to asymptotically. The latest attached brief should be treated as the status authority: an older August 19 repository note still records odd depths 3 and 5 as open, whereas the newer brief states that all depths through 6 are now closed. [filecite turn0file0](#)

The most promising new direction I find is not another brute-force certificate. Starting from the repository's exact finite-sum formula, the depth dependence can be reorganized into an **alternating binomial transform of a normalized sequence** $M_t^{(r)}$, with $r = j - 1$. In the known scaling limit, that normalized sequence collapses to the geometric moment sequence x^t , so the entire knife becomes exactly $(1 - x)^r$, reproducing the observed odd/even parity law. This suggests looking for a positive integral representation of the finite- λ sequence—a finite-dimensional analogue of the Hausdorff-moment mechanism that Mansfield successfully uses for Gregory coefficients in his Veneziano-positivity proof. Mansfield's proof is particularly instructive because it combines three ingredients that resemble what is needed here: a representation change by adding an object with zero residue, a positive moment representation on $[0, 1]$, and induction in the partial-wave index. ¹

A second exact observation comes with essentially no computational cost. After the same reorganization, the pure depth kernel is

$$B_{r,t} = r^t (H - r)^t = t!^2 \binom{r}{t} \binom{H - r}{t}, \quad H = \frac{D + 4n - 7}{2}.$$

Its adjacent 2×2 minors satisfy the exact factorization

$$\det \begin{pmatrix} B_{r,t} & B_{r,t+1} \\ B_{r+1,t} & B_{r+1,t+1} \end{pmatrix} = B_{r,t} B_{r+1,t+1} (H - 2r - 1).$$

Since $r \leq n - 2$,

$$H - 2r - 1 \geq \frac{D - 1}{2} > 0$$

throughout the physical domain $D > 3$. Thus every defined adjacent 2×2 minor of this *j-direction kernel* is strictly positive. This does not prove the theorem, and it should not be confused with the already-rejected total-positivity conjectures in the coefficient index m ; but it is the first concrete evidence here for a sign-regular/total-positive structure specifically in the previously underexplored depth index. The algebra follows directly from the exact closed form recorded in the repository.

The broader research portfolio remains worthwhile. In technology, the strongest alternative is verification-first AI for scientific computation; in business, measuring generative-AI productivity in small creative software teams; in health, distinguishing causal effects from selection in adolescent problematic gaming; in environment, the electricity and grid implications of AI data centres; in policy, the now-operative EU AI Act Article 50 transparency rules; and in education, identifying when AI tutoring improves learning rather than merely task completion. OECD data show AI use by firms reached 20.2% in 2025 and 57.3% in ICT; WHO/ Europe reports 11% problematic social-media behavior and 12% at risk of problematic gaming in its 2022 adolescent survey; the IEA projects data-centre electricity consumption around 945 TWh in 2030; and EU Article 50 transparency obligations have applied since August 2, 2026. ²

Priority decision requested: which track should ultimately dominate the research program—technology, business, health, environment, policy, education, or the attached mathematical keystone? In the absence of another choice, the rest of this report prioritizes the **keystone theorem**.

Scope and Assumptions

The initial conditions are only partly unspecified. The attachment materially resolves the subject, while audience, budget, institutional constraints, and final publication deadline remain open.

Item	Working assumption
Specific subject	The attached all-orders Jacobi/partial-wave positivity problem is the default primary subject. filecite turn0file0
Mathematical scope	All $n \geq 3$, $3 \leq j \leq n - 1$, $\lambda > 0$, and $3 < D \leq \widehat{T}(\lambda)$, with boundary nonnegativity rather than unjustified strict positivity. filecite turn0file0
Audience	Research-level mathematical physics / special functions / symbolic computation; unspecified by the request.
Deadline	Unspecified; project-duration estimates below are planning estimates rather than commitments.
Budget	Unspecified; assume one primary researcher, access to the existing exact-arithmetic codebase, ordinary workstation-class computing, and occasional specialist review.
Deliverable	This analytical report is specified; whether the ultimate deliverable is a paper, formal proof, certified artifact, or all three remains unspecified.

Item	Working assumption
Evidence standard	Exact algebra, exact rational arithmetic, certified inequalities, or established theorems. Numerical sweeps are for falsification and discovery, not final proof. This is consistent with the project's own error history and methodology.
Success criterion	Best case: an all- j theorem. Acceptable intermediate success: a uniform recurrence or integral representation reducing the theorem to finitely many base inequalities. A sharp counterexample to a proposed uniform mechanism is also valuable.

The core mathematical object is a nonnegative polynomial $F(u)$ of degree $n - 1$ expanded in Jacobi polynomials $P_m^{(-1/2, D/2-2)}(1 - 2u)$. With $m = n - j$, the signed coefficient $(-1)^m f_m$ is called **knife $_j$** , and the conjectured physical region is bounded by

$$\widehat{T}(\lambda) = \min_{k \in \mathbb{Z}, k \geq 3} T_k(\lambda),$$

$$T_k(\lambda) = \frac{3(2k-3)}{k(k-2)} [\lambda^2 + (2k-2)\lambda + 1] + 2k.$$

The brief records the key empirical/theoretical dichotomy: odd j appears to have no finite positivity threshold, while even j becomes marginal against the shore. [filecite turn0file0](#)

The fixed- j scaling limit has already been derived in closed form. Taking $n = \rho\lambda$, $D = d\lambda$, and $\lambda \rightarrow \infty$,

$$\text{knife}_j \rightarrow \left(\frac{2\rho^2 + 12\rho + 6 - d\rho}{6(\rho + 1)^2} \right)^{j-1}.$$

For odd j , $j - 1$ is even, so the leading expression is nonnegative irrespective of the sign of the bracket. For even j , the bracket changes sign at

$$d = 2\rho + 12 + \frac{6}{\rho},$$

whose minimum occurs at $\rho = \sqrt{3}$ and equals $12 + 4\sqrt{3}$. The repository also records positive subleading behavior at the tangency for the explicitly checked even knives.

This scaling result is powerful but explicitly **not uniform in j** : its derivation fixes j before $\lambda \rightarrow \infty$, and the convergence of the central-factorial coefficients slows as the summation order grows. That limitation is exactly why simply extending the asymptotic calculation does not solve the all-depth theorem.

There is also a classical dimension direction that is already structurally understood. Gneiting's review of Schoenberg-type positive-definite functions on spheres describes the nested dimension classes and dimension walks for Gegenbauer expansions; this is the mathematical background of the project's descent step. ³ The attachment therefore correctly identifies the remaining missing direction as **depth j** rather than dimension D . [filecite turn0file0](#)

Finally, negative results in the repository materially constrain this research. The project has already invalidated endpoint-minimum claims, log-concavity claims, naive saddle-point reductions, low-spin

dominance, fixed-window uniformity, and several implementations that looked certified until independently tested. In particular, the ERRATA records explicit counterexamples to the endpoint/log-concavity route and emphasizes that grids must track moving boundaries rather than convenient absolute coordinates. Those routes should remain closed unless a genuinely different quantity is being studied.

Cross-Domain Research Portfolio

Technology — verification-first AI for computational mathematics and scientific software. The attached project itself provides a strong case study: independent verification uncovered bugs that self-checks missed, and several mathematically attractive statements were retracted after exact counterexamples. A rigorous technology study would ask which architecture best separates conjecture generation, symbolic derivation, exact evaluation, and adversarial verification; how often independent implementations catch correlated errors; and whether automatically generated metamorphic tests can identify vacuous checks. The methodology should use the existing ScienceBro history as a retrospective dataset, deliberately inject mutations into proof/certificate code, and compare self-checking, independent exact engines, formal proof assistants, and differential testing. Primary evidence should be repository commits/artifacts, FLINT-based exact computations, and formal-verification literature rather than vendor benchmarks. The high-value deliverable is a reusable verification protocol for AI-assisted scientific programming, not another coding benchmark. [filecite.bturn0file0](#)

Business — where generative AI actually changes productivity in creative technical teams. OECD data show a large and fast-growing adoption gap across firm sizes: in 2025, 52.0% of large firms but 17.4% of small firms in covered OECD countries reported AI use, while ICT had the highest industry adoption at 57.3%. ⁴ Existing field evidence also warns against assuming a uniform productivity effect. Brynjolfsson, Li, and Raymond found an average 14% productivity increase among 5,179 customer-support agents, concentrated heavily among less experienced workers; a later field experiment across 66 firms and 7,137 knowledge workers found meaningful reductions in email time but not broad shifts in task composition. ⁵ A useful game/creator-oriented study would ask whether AI reduces prototype cycle time, bug-resolution time, art/content iteration cost, and administrative work **without degrading shipped quality or increasing rework**. The preferred methodology is a within-team or randomized crossover design using repository telemetry, issue trackers, asset revision counts, independent quality ratings, and delayed defect rates rather than self-reported “time saved.”

Health — problematic gaming and adolescent mental health: causal pathway or risk marker? WHO/Europe's HBSC survey of almost 280,000 adolescents found 12% classified as at risk of problematic gaming and 11% showing problematic social-media behavior, but WHO also stresses that heavy digital use is not synonymous with problematic use and can coexist with social benefits. ⁶ Recent longitudinal work is more informative than cross-sectional alarmism: a 2025 Tokyo Teen Cohort analysis of 3,171 adolescents found problematic online gaming at age 14 associated with elevated subsequent risks of depression, anxiety, psychotic experiences, and reduced well-being at age 16 after extensive adjustment, while a 2026 ABCD analysis found prospective associations between problematic screen-use measures and several mental and behavioral outcomes. ⁷ The research questions should separate ordinary gaming from loss-of-control/problematic behavior, identify confounding and bidirectionality, and estimate which symptoms are plausible intervention targets. Methodologically, the right project is a preregistered systematic review emphasizing longitudinal cohorts and interventions, followed—if data permit—by individual-level causal modeling. It should not collapse “hours played” into “addiction.”

Environment — AI/data-centre electricity demand, grid constraints, and flexible computing. The IEA's current base case projects global data-centre electricity consumption to more than double to roughly 945 TWh by 2030, with AI the most important driver of the growth; it also expects renewables, gas, nuclear, storage, and grid investment all to play roles in supplying it. ⁸ The central questions are not simply “how much electricity does AI use?” but where incremental load appears, what marginal generation serves it, whether training/inference can shift temporally, and whether data centres can provide demand flexibility rather than acting as rigid loads. Recent electrical-engineering work is actively examining grid-aware data-centre operation. ⁹ A medium project would combine IEA scenarios with ENTSO-E/national grid data and hourly carbon-intensity modeling; a serious long project should model connection queues, regional congestion, storage, and workload-shifting under alternative AI-demand scenarios.

Policy — operational EU AI Act compliance for creators, games, and interactive AI. This is particularly time-sensitive because Article 50 transparency obligations began applying on **August 2, 2026**, and the Commission and national authorities have begun enforcement. The Commission states that covered interactive AI systems must disclose that users are interacting with AI, while certain AI-generated or manipulated outputs face marking or disclosure obligations; limited grandfathering for some pre-existing systems runs to December 2, 2026. ¹⁰ A useful study would map concrete creator/game workflows—NPC dialogue, generated art/audio, marketing assets, user-generated AI content, moderation tools, synthetic voice/video—to provider/deployer roles and Article 50 duties. The methodology should begin with the Regulation and July 2026 Commission guidance, construct scenario matrices, and have the legal interpretation reviewed by qualified EU counsel before being used operationally. The deliverable would be a product-level compliance decision tree rather than a generic AI Act summary.

Education — when generative AI improves programming learning instead of bypassing it. UNESCO's guidance calls for human-centred use, pedagogical validation, privacy protection, and preservation of learner agency rather than unstructured deployment. ¹¹ The evidence base is now moving beyond opinion. A 2025 meta-analysis found 11 randomized trials of generative-AI teaching in medical education, while an August 2026 NBER field experiment with more than 6,000 middle-school students found that AI support's value depended on the instructional structure: it improved recovery after mistakes, while the most encouraging delayed-learning result appeared when AI was integrated into a mastery workflow. ¹² A game-development/programming version should therefore measure delayed unaided transfer, debugging ability, code explanation, and novel-problem performance—not only assignment completion. The clean experiment is AI assistance versus structured AI tutoring versus no AI, with identical final unaided assessments.

The expected research outputs and resource profiles compare as follows. These are planning estimates because budget and deadline are unspecified.

Track	Short project	Medium project	Long project	Estimated resources
Technology	3–5 days: error taxonomy and verification architecture from the existing repository	3–4 weeks: mutation-testing benchmark of independent verifiers	8–12 weeks: reproducible benchmark suite and paper	1 research engineer; exact arithmetic; optional Lean/formal-methods specialist

Track	Short project	Medium project	Long project	Estimated resources
Business	3–5 days: evidence review + measurement protocol	4–6 weeks: instrument one small technical team	10–16 weeks: controlled multi-team study with quality outcomes	1 analyst; engineering telemetry; 2–5 cooperating teams
Health	5–7 days: rapid evidence map	4–6 weeks: preregistered systematic review/meta-analysis	3–6 months: longitudinal causal reanalysis or prospective study	Epidemiologist/statistician; PubMed/ Cochrane; ethics/ data access for primary research
Environment	3–5 days: scenario synthesis from IEA/grid data	3–5 weeks: hourly regional electricity/carbon model	8–16 weeks: grid-constrained flexibility study	Energy analyst; Python/R; IEA + grid datasets
Policy	2–4 days: Article 50 creator/game obligations map	2–3 weeks: product scenario matrix and gap analysis	6–10 weeks: compliance playbook reviewed against enforcement guidance	EU policy/legal researcher; qualified legal review for operational use
Education	3–5 days: curriculum/ evidence review	4–6 weeks: small controlled pilot	10–16 weeks: preregistered delayed-transfer RCT	Instructor/ researcher; 30–100+ learners for pilot; ethics review where applicable
Keystone mathematics	1–3 days: exact falsification of the new t -kernel hypotheses	2–5 weeks: derive uniform determinant/measure/ recurrence lemma	6–12+ weeks: all-depth theorem, independent verification, manuscript	1 mathematical physicist + exact engine; special-functions/total-positivity review highly valuable

Recommended Immediate Topic: All-Order Jacobi Positivity

The reason to prioritize the keystone problem is not merely that it is the attached topic. It has already crossed the stage where more blind experimentation is useful. The low-depth theorem machinery, exact closed form, scaling limit, shore geometry, parity behavior, dimension walk, and failure modes collectively constrain the missing theorem enough that a **structural attack in the depth index** is now realistic.

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What is genuinely established

The project reports exact Bernstein-based closure through depth 6, with both parities covered by the later construction, as well as a separate exact certification of monotonicity in D over the relevant range. The attached brief explicitly labels the infinite-depth family as open. [filecite turns file](#) The older unglued-keystone note documents how this position was reached: the proof architecture had to be separated into continuum, integrality, and monotonicity components, and several bugs were found by probing points or branches that previous self-checks did not cover.

The scaling theorem is considerably stronger than a numerical pattern: from an exact finite expression,

$$\text{knife}_j \sim \sum_{t=0}^{j-1} (-1)^t E_{2t}(n) s^{-2t} R_t, \quad s = n + \lambda - 1,$$

with

$$R_t = \frac{\prod_{i=1}^t (j-i) \prod_{i=1}^t (D + 4n - 2j - 5 - 2(i-1))}{\prod_{i=1}^t (n-i) \prod_{i=1}^t (2n-1-2i)},$$

the leading large- λ terms become a binomial series and sum by Newton's theorem to the $(1-x)^{j-1}$ structure above. The repository reports exact comparisons against the reference engine for multiple knife orders and a prediction/check at $j = 6$.

That means the research question can be sharpened considerably:

What finite- (n, λ, D) structure deforms the exact asymptotic binomial identity without destroying positivity on the shore?

That is more useful than asking vaguely whether the Jacobi coefficients are positive.

An exact depth-kernel reorganization

Set

$$r = j - 1, \quad H = \frac{D + 4n - 7}{2}, \quad s = n + \lambda - 1.$$

The two j -dependent products in the exact formula become

$$\prod_{i=1}^t (j-i) = r^t,$$

and

$$\prod_{i=1}^t (D + 4n - 2j - 5 - 2(i-1)) = 2^t (H - r)^t.$$

Meanwhile,

$$\prod_{i=1}^t (2n - 1 - 2i) = 2^t \left(n - \frac{3}{2} \right)^{\underline{t}}.$$

The factors of 2^t cancel. Therefore, up to the already-known positive normalization relating the closed sum to the knife sign,

$$K_r = \sum_{t=0}^r (-1)^t c_t r^{\underline{t}} (H - r)^{\underline{t}}$$

with

$$c_t = \frac{E_{2t}(n)}{s^{2t} (n-1)^{\underline{t}} \left(n - \frac{3}{2} \right)^{\underline{t}}}.$$

This is just a rearrangement of the repository's exact formula, not an asymptotic approximation.

Using $r^{\underline{t}} = t! \binom{r}{t}$, define

$$M_t^{(r)} = t! c_t (H - r)^{\underline{t}}.$$

Then

$$K_r = \sum_{t=0}^r (-1)^t \binom{r}{t} M_t^{(r)}.$$

Equivalently, in forward-difference notation,

$$K_r = (-1)^r \Delta^r M_0^{(r)}.$$

This is, in my view, the most important reparametrization to test next. It converts “all depths” into a problem about a **single normalized sequence in the summation index t** .

Most importantly, the exact scaling theorem says that in the scaling regime

$$M_t^{(r)} \longrightarrow x^t, \quad x = \frac{\rho(d + 4\rho)}{6(\rho + 1)^2},$$

because then

$$K_r \longrightarrow \sum_{t=0}^r (-1)^t \binom{r}{t} x^t = (1 - x)^r.$$

Thus the normalized finite sequence $M_t^{(r)}$ is precisely the object whose large-scale limit becomes a one-point moment sequence.

The positive-measure hypothesis

This motivates a very specific candidate representation:

$$M_t^{(r)} = \int x^t d\nu_r(x), \quad d\nu_r(x) \geq 0.$$

If such a representation exists, then automatically

$$K_r = \int (1-x)^r d\nu_r(x).$$

This form would explain the observed parity law almost embarrassingly well.

When j is odd, $r = j - 1$ is even, so $(1-x)^r \geq 0$ for every real x . A positive real-line measure would immediately imply the empirical statement that odd knives never cut. When j is even, r is odd, so support crossing $x = 1$ can make the integral negative. If, however, the physical-shore condition forced $\text{supp } \nu_r \subseteq [0, 1]$, then every even knife would also be nonnegative below the shore. In the scaling limit the measure would simply collapse to δ_x , with the critical curve corresponding to $x = 1$. This matches both the parity structure and the tangency mechanism already proved asymptotically.

This proposal must be distinguished from the dead “Stieltjes moment structure” route mentioned in the brief. The candidate here is **not** a moment hypothesis for the Jacobi coefficient sequence $m \mapsto f_m$; it is a moment representation for the newly normalized **internal summation sequence** $t \mapsto M_t^{(r)}$ after extracting the exact binomial depth kernel. The distinction is substantive. Nevertheless, the first action should be to check the old experimental logs: if the rejected moment experiment already tested exactly this normalized $M_t^{(r)}$, the route should be killed immediately rather than renamed. The brief does not specify that normalization, so the overlap cannot be established from the supplied materials.

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The literature makes this proposal less speculative than it might appear. Mansfield's 2026 proof uses coefficients G_k with an explicit positive $[0, 1]$ -measure representation,

$$G_k = \int_0^1 x^{k-2} w(x) dx,$$

and uses that representation to turn polynomial positivity on $[0, 1]$ into positivity of linear combinations of the G_k . He also uses the corresponding positive Hankel matrices in the proof. ¹ The truncated moment problem has a mature algebraic formulation in terms of positive moment and localizing matrices; Curto and Fialkow give a general truncated K -moment framework through positivity and extensions of the Riesz functional. ¹³

This candidate is therefore highly falsifiable. For each exact physical tuple, build the finite moment matrices

$$\mathcal{H}_q = (M_{a+b}^{(r)})_{a,b=0}^q,$$

together with the $[0, 1]$ -support localizers built from $M_{k+1}^{(r)}$ and $M_k^{(r)} - M_{k+1}^{(r)}$, whenever the available moment order permits them. A single negative exact principal minor kills the naive positive-measure hypothesis for that region. Systematic positivity, especially if the $LDL^\top$ pivots factor, would supply the next theorem target. Moment-matrix and flat-extension techniques are standard tools for such truncated support questions. 14

An exact depth recurrence inside the normalized sequence

The normalization has another useful property. Since only $(H - r)^t$ depends on r inside $M_t^{(r)}$,

$$\frac{M_t^{(r+1)}}{M_t^{(r)}} = \frac{(H - r - 1)^t}{(H - r)^t} = \boxed{\frac{H - r - t}{H - r} = 1 - \frac{t}{H - r}}.$$

Thus

$$\boxed{M_t^{(r+1)} = \left(1 - \frac{t}{H - r}\right) M_t^{(r)}}.$$

This is an **exact depth recursion**, not a fitted law. It says that incrementing depth acts on the normalized t -sequence by multiplication with an affine function of t . That is precisely the kind of low-complexity structure that could support an induction or a measure-transform argument. It is obtained algebraically from the same exact finite-sum formula.

A successful theorem would therefore not need to rediscover each knife independently. It could prove that a suitable class of sequences or representing measures is preserved under the transformation

$$M_t \mapsto \left(1 - \frac{t}{A}\right) M_t, \quad A = H - r,$$

while the physical parameter inequalities guarantee A remains in the required range.

The depth kernel has unexpectedly clean minors

Strip off c_t and consider

$$B_{r,t} = r^t (H - r)^t = t!^2 \binom{r}{t} \binom{H - r}{t}.$$

Its ratio in the column direction is

$$\frac{B_{r,t+1}}{B_{r,t}} = (r - t)(H - r - t).$$

Consequently,

$$\begin{aligned}
& \frac{B_{r+1,t+1}}{B_{r+1,t}} - \frac{B_{r,t+1}}{B_{r,t}} \\
&= (r+1-t)(H-r-t-1) - (r-t)(H-r-t) \\
&= H-2r-1.
\end{aligned}$$

Therefore

$$\det \begin{pmatrix} B_{r,t} & B_{r,t+1} \\ B_{r+1,t} & B_{r+1,t+1} \end{pmatrix} = B_{r,t} B_{r+1,t} (H-2r-1).$$

At the greatest allowed $r = n-2$,

$$H-2r-1 = \frac{D-1}{2} > 0.$$

So every adjacent 2×2 minor that exists on the physical triangular index domain is positive.

That is a meaningful structural clue because discrete Hahn-type systems naturally organize factorial/binomial kernels, and the DLMF Hahn class is built around difference operators and dualities rather than differential operators. ¹⁵ I would **not** yet claim that B is totally positive of all orders; the next exact experiment is to factor solid minors of orders 3, 4, . . . and determine whether there is a uniform product formula. The point is that this is now a precise theorem candidate in the j -direction, not an analogy.

The rejected total-positivity work in the project should therefore not be casually resurrected. The relevant distinction is that the failed work concerned other coefficient/index structures, while $B_{r,t}$ is a newly exposed kernel whose row variable **is the depth** r . The test is cheap enough that there is no reason to debate the distinction philosophically: calculate exact higher minors and either keep or discard it.

Concrete Proof Program and Literature

Three routes deserve active investigation, in this order.

Primary route: normalized t -sequence $\rightarrow$ positive measure $\rightarrow$ all depths. The reason for ranking this first is explanatory power. It reproduces the exact asymptotic binomial law, naturally explains the odd/even split, converts the theorem into positivity of $(1-x)^r$, and has a direct precedent in Mansfield's use of a Hausdorff moment sequence. ¹ It also makes the desired integral representation explicit rather than merely hoping one exists.

The first falsification suite should evaluate $M_t^{(r)}$ using exact rationals for all already-certified depths and many higher, concentrating on the shore rather than on arbitrary D . The repository's own ERRATA demonstrates why relative-to-boundary sampling is essential: earlier conjectures survived because tests sat deep inside a moving physical region rather than near the boundary that determined the sign. Parameters should therefore include exact $D = T_k(\lambda)$ values, nearby admissible D , critical-curve points, very small and very large λ , j both fixed and proportional to n , and points around any parity-transition geometry.

A strong positive signal would not merely be “all tested Hankel minors are positive.” It would be **factorization of the pivots into known positive factors**: powers of s , falling factorials with manifest signs, $D - 3$, shore slack $T_k - D$, or the conic critical polynomial already present in the project. The decisive target is an algebraic $LDL^\top$ decomposition or explicit beta-mixture measure.

There is a striking literature match here. DLMF records the Askey–Gasper positive-sum theorem for Jacobi polynomials,

$$\sum_{m=0}^n \frac{(\eta+1)_{n-m}}{(n-m)!} \frac{(\eta+1)_m}{m!} \frac{P_m^{(\alpha,\beta)}(x)}{P_m^{(\beta,\alpha)}(1)} \geq 0,$$

under $\alpha + \beta \geq \eta \geq 0$ and $\beta \geq -1/2$. ¹⁶ In the present problem the Jacobi parameters are

$$\alpha = -\frac{1}{2}, \quad \beta = \frac{D}{2} - 2,$$

so the basic requirement $\beta \geq -1/2$ coincides exactly with $D \geq 3$. The remaining hypotheses and, crucially, the weights do **not** match the keystone problem, so the Askey–Gasper inequality is not the missing theorem. But it shows that positive weighted Jacobi sums at precisely this endpoint parameter are part of a well-developed structure. ¹⁶

Koornwinder's 1978 paper is even more conceptually relevant: it derives nonnegative linearization coefficients from addition formulas and gives sufficient conditions for nonnegative **connection coefficients** between related orthogonal systems, with Jacobi polynomials among the applications. ¹⁷ DLMF's Jacobi connection formula likewise exposes Pochhammer factors whose signs can become manifest in suitable parameter orderings. ¹⁸ This supports a focused literature task: try to identify the finite $M_t^{(r)}$ sequence as moments or connection coefficients of an auxiliary Jacobi/Hahn system rather than attacking the original Jacobi integral directly.

Secondary route: a telescoping “null corrector” in the finite t -sum. Mansfield's most transferable idea is not his particular large- D Laurent series—which the project's own analysis finds unsuitable in this regime—but the act of **changing the representation by adding zero**. In Mansfield's ten-dimensional proof, a rational term with vanishing residue is added to cancel finitely many bad Laurent coefficients, after which positivity is propagated by induction in j . ¹⁹

The exact finite-sum analogue is straightforward. Write

$$a_{r,t} = (-1)^t \binom{r}{t} M_t^{(r)}.$$

Seek a rational certificate $G_{r,t}$ such that

$$G_{r,0} = G_{r,r+1} = 0$$

and

$$\boxed{w_{r,t} = a_{r,t} + G_{r,t+1} - G_{r,t} \geq 0}$$

termwise throughout the shore region. Telescoping gives

$$\sum_{t=0}^r w_{r,t} = \sum_{t=0}^r a_{r,t} = K_r.$$

A practical low-complexity ansatz is

$$G_{r,t} = t(r+1-t) a_{r,t} R(t; r, n, \lambda, D),$$

where R is a rational function of low degree and the prefactor automatically enforces endpoint vanishing. The existing exact depths can be used to solve for or guess R . The falsification criterion is severe: if the minimum degree needed for R grows steadily with r , this is probably another disguised depth-by-depth certificate and should be abandoned. A depth-independent or bounded-degree R , however, would be a genuine keystone.

This is also a natural place for creative telescoping. Modern symbolic-summation methods are explicitly designed to obtain parameter recurrences for definite hypergeometric-type sums, although the central-factorial $E_{2t}(n)$ factor means the exact summand must first be placed in the right recurrence framework.

²⁰ The aim should not be “ask a CAS to prove it,” but derive a short recurrence whose coefficients can then be checked for sign by exact algebra.

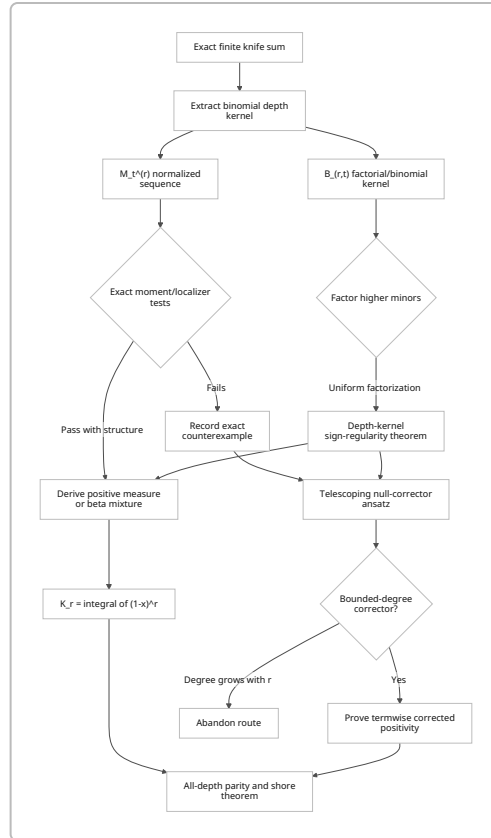
Tertiary route: prove sign regularity or total positivity of the depth kernel and combine it with a structured c_t . The positive adjacent minors derived above justify a serious higher-minor test. The appropriate target is not “ B seems TP numerically,” but an exact determinant identity for

$$\det[B_{r_a, t_b}]_{a,b=1}^q$$

or, initially, all solid minors. The binomial-product form suggests looking at Hahn/dual-Hahn determinant identities and nonintersecting-path interpretations. DLMF’s Hahn class and dualities provide the closest standard discrete-orthogonal-polynomial framework. ¹⁵

Total positivity by itself will not finish the theorem: the input vector carries the alternating factor $(-1)^t$, so a variation-diminishing theorem must be paired with additional structure of c_t or M_t . The point of the kernel program is to identify exactly what additional structure is required. A useful endpoint would be a theorem of the form “the checkerboard-signed depth transform maps class $\mathcal{C}$ of normalized central-factorial sequences into nonnegative sequences,” with $c_t \in \mathcal{C}$ proved separately.

The following diagram shows how these routes should interact rather than proceed as three independent brute-force projects:



Two nearby strands of literature should be read with sharply different expectations. Gneiting/Schoenberg gives a dependable **dimension walk**, but does not supply the missing j -direction theorem. ³ Mansfield proves all partial waves for the ordinary Veneziano problem through $D \leq 10$, but his original Laurent-coefficient positivity already becomes increasingly difficult above $D = 6$; he repairs $D = 10$ by a null-residue corrector rather than by extending raw coefficient positivity. ²¹ The project's brief is therefore right that his theorem does not simply contain the present large- D , λ -deformed problem.

What should transfer is the **architecture**: find the representation in which a positive moment sequence and an order recursion become visible.

The underlying amplitude family itself comes from Cheung, Hillman, and Remmen's closed-string bootstrap deformation, in which level truncation and crossing yield a λ -dependent spectrum and hypergeometric amplitude; the Virasoro-Shapiro point is recovered at $\lambda = 1$. ²² That source is useful for ensuring that future mathematical transformations preserve the actual amplitude parameters rather than artifacts of one coefficient engine.

Deliverables, Timeline, and Prioritized Next Steps

For the recommended mathematical topic, I would structure the project around **decision gates**, with every early stage capable of killing a bad idea before expensive certification work begins.

Horizon	Research task	Concrete deliverable	Go/no-go condition	Effort
Short	Derive $M_t^{(r)}$, moment/localizer matrices, and higher B -kernel minors in exact arithmetic	A compact JSON/Markdown artifact over adversarial physical points; symbolic factorizations through moderate orders	Continue moment route only if no exact localizer failure at certified depths and higher-depth stress points; continue kernel route only if minors show a stable sign/factor pattern	~1–3 researcher-days
Short	Search bounded-degree telescoping correctors on depths already proved	Exact candidate $G_{r,t}$ or proof that low-degree ansätze fail	Continue only if one functional form works across several depths without retuning degree	~2–4 researcher-days
Medium	Turn the strongest observed pattern into a parameter-uniform lemma	One self-contained theorem: positive measure, determinant factorization, or positivity-preserving depth transform	Must remove j from the computational certificate count	~2–5 weeks
Medium	Independent adversarial verifier	Second implementation using the exact reference engine but not importing prover constructions	Every claimed algebraic identity must survive independent boundary and outside-window tests	Parallel, several person-days
Long	Close remaining parameter regions and combine with dimension/integrality pieces	All- n, j, λ, D theorem or a finite family of base cases plus uniform induction	No measured step may remain inside the logical chain	~6–12+ weeks, highly uncertainty-dependent
Long	Publication package	Proof manuscript, machine-readable certificates/tests, explicit errata/status audit, reproducibility instructions	Independent reproduction from clean checkout	Additional ~2–4 weeks after theorem structure stabilizes

The priority order should be:

First, test the normalized t -sequence rather than a new Jacobi-coefficient conjecture. Implement

$$M_t^{(r)} = \frac{t! E_{2t}(n)(H-r)^t}{s^{2t}(n-1)^t(n-\frac{3}{2})^t}$$

directly from the closed form, and compare its reconstructed binomial transform against the reference knife engine at points where the knife is positive, zero, and negative. This prevents a normalization or off-by-one error from contaminating the whole program; the project's own ERRATA shows that such checks are indispensable.

Second, run three adversarial tests on $M_t^{(r)}$: ordinary Hankel positivity; $[0, 1]$ -support localizers specifically at the shore; and the same tests deliberately above the shore. The strongest possible outcome is not just shore positivity but the qualitative pattern “positive real-line moment structure persists for odd knives while the $[0, 1]$ support condition breaks when even knives cross their threshold.” That would reproduce the observed physics mechanism at the level of exact algebra. The Mansfield precedent shows that a Hausdorff-moment representation can be strong enough to drive an order induction rather than merely decorate it. ¹

Third, factor depth-kernel minors independently of the moment test. The universal 2×2 factor $H - 2r - 1$ is too clean to ignore. Test solid minors through as high an order as exact polynomial arithmetic remains trivial, then search the Hahn literature only after the pattern is known. This avoids the common failure mode of forcing the object into a famous theorem whose hypotheses do not actually match.

Fourth, transpose Mansfield's null-corrector idea only if the direct measure route stalls. Do not retry his large- D Laurent expansion. Instead, exploit the finite sum's own null space: a telescoping difference $G_{t+1} - G_t$ changes termwise signs without changing the knife. Mansfield's successful proof establishes that “add zero to reveal positivity” is a legitimate structural technique, and that an induction in the angular-momentum index can then become possible. ¹⁹

Fifth, keep the existing certified low depths as base cases, not as the main method. If a uniform recursion begins only at some j_0 , the exact depth- ≤ 6 results are already positioned to supply the base. ^{filecite turn0file0} The objective should therefore be a recurrence or integral transform that starts no later than the existing frontier, rather than a beautiful induction whose base case lies at depth 50.

Sixth, do not pursue the empirical gap constant $C \simeq 2.398$ as the main theorem target. The brief correctly labels the apparent linear gap law as measured rather than proved. ^{filecite turn0file0} It may eventually emerge as a finite- λ property of the support edge or first moment of ν_r , but proving the full positivity theorem does not require first identifying whether that constant is $12/5$, another closed form, or not universal at all.

Primary-source reading list

Source	Why it should be read now
Cheung, Hillman & Remmen, <i>Uniqueness Criteria for the Virasoro-Shapiro Amplitude</i>	Primary source for the λ -deformed closed-string amplitude family and its bootstrap origin. ²²

Source	Why it should be read now
Mansfield, <i>Positivity of the Veneziano Amplitude in Ten Dimensions</i>	Closest proof architecture: null-residue corrector, j -induction, and an explicit Hausdorff-moment sequence. ¹⁹
Gneiting, <i>Strictly and non-strictly positive definite functions on spheres</i>	Dimension-walk/Schoenberg background and the distinction between dimension structure and the missing depth structure. ³
NIST DLMF §18.14, Jacobi positive sums	Authoritative statement of Askey–Gasper-type positive Jacobi sums; parameter $\beta \geq -1/2$ aligns directly with $D \geq 3$. ¹⁶
NIST DLMF §18.18, Jacobi connection formulas	Explicit positive/negative Pochhammer structure for basis changes; useful when searching for a positivity-preserving representation. ¹⁸
Koornwinder, <i>Positivity Proofs for Linearization and Connection Coefficients...</i>	Classical mechanism converting addition formulas into nonnegative linearization/connection coefficients. ¹⁷
DLMF §18.19–§18.21, Hahn class and dualities	Closest standard framework for the newly exposed factorial/binomial depth kernel. ¹⁵
Curto & Fialkow, <i>An analogue of the Riesz–Haviland theorem for the truncated moment problem</i>	Rigorous framework for turning finite moment/localizing-matrix evidence into a representing-measure theorem. ¹³
European Commission, Article 50 transparency FAQ	Primary source for the policy-track alternative and the rules applying from August 2, 2026. ²³
OECD, 2026 AI adoption data	Primary statistical source for the business-track alternative. ⁴
WHO/Europe, <i>Teens, screens and mental health</i>	Primary HBSC source for the health-track alternative. ⁶
IEA, <i>Energy and AI executive summary</i>	Primary scenario source for the environment-track alternative. ⁸
UNESCO, <i>Guidance for generative AI in education and research</i>	Baseline policy/pedagogical framework for the education-track alternative. ²⁴

The central research bet is therefore quite specific: **the finite- λ theorem is likely hiding in the normalized t -kernel, not in another direct inequality for the original Jacobi coefficient.** The exact binomial reorganization, the affine depth update $M_t^{(r+1)}/M_t^{(r)} = 1 - t/(H - r)$, the strictly positive adjacent minors of the factorial depth kernel, the asymptotic collapse $M_t \rightarrow x^t$, and Mansfield's successful use of moment structure plus a null corrector all point to the same place. ¹

That route has the right scientific property for this project: it can fail **quickly and exactly**. If the relevant moment/localizing matrix develops a negative rational minor at depth 7, the hypothesis dies immediately. If

it survives and the pivots factor, there is a plausible path from the already-proved finite depths to the first genuinely uniform theorem in j .

1 19 21 Positivity of the Veneziano Amplitude in Ten Dimensions

<https://arxiv.org/html/2502.20372v2>

2 4 <https://www.oecd.org/en/about/news/announcements/2026/01/ai-use-by-individuals-surges-across-the-oecd-as-adoption-by-firms-continues-to-expand.html>

<https://www.oecd.org/en/about/news/announcements/2026/01/ai-use-by-individuals-surges-across-the-oecd-as-adoption-by-firms-continues-to-expand.html>

3 Strictly and non-strictly positive definite functions on spheres

<https://arxiv.org/abs/1111.7077>

5 <https://www.nber.org/papers/w31161>

<https://www.nber.org/papers/w31161>

6 <https://www.who.int/europe/news-room/25-09-2024-teens--screens-and-mental-health>

<https://www.who.int/europe/news-room/25-09-2024-teens--screens-and-mental-health>

7 <https://pubmed.ncbi.nlm.nih.gov/40759741/>

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8 <https://www.iea.org/reports/energy-and-ai/executive-summary>

<https://www.iea.org/reports/energy-and-ai/executive-summary>

9 <https://www.nature.com/articles/s44287-025-00255-6>

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10 <https://ai-act-service-desk.ec.europa.eu/en/ai-act/faq/when-does-enforcement-start>

<https://ai-act-service-desk.ec.europa.eu/en/ai-act/faq/when-does-enforcement-start>

11 24 <https://www.unesco.org/fr/digital-education/ai-future-learning/guidance>

<https://www.unesco.org/fr/digital-education/ai-future-learning/guidance>

12 <https://pubmed.ncbi.nlm.nih.gov/40830476/>

<https://pubmed.ncbi.nlm.nih.gov/40830476/>

13 An analogue of the Riesz–Haviland theorem for the truncated moment problem - ScienceDirect

https://www.sciencedirect.com/science/article/pii/S0022123608003340?utm_source=chatgpt.com

14 Centrum Wiskunde & Informatica: A sparse flat extension theorem for moment matrices

https://ir.cwi.nl/pub/13265?utm_source=chatgpt.com

15 <https://dlmf.nist.gov/18.19>

<https://dlmf.nist.gov/18.19>

16 <https://dlmf.nist.gov/18.14>

<https://dlmf.nist.gov/18.14>

17 <https://londmathsoc.onlinelibrary.wiley.com/doi/abs/10.1112/jlms/s2-18.1.101>

<https://londmathsoc.onlinelibrary.wiley.com/doi/abs/10.1112/jlms/s2-18.1.101>

18 <https://dlmf.nist.gov/18.18>

<https://dlmf.nist.gov/18.18>

²⁰ <https://arxiv.org/abs/0808.2596>

<https://arxiv.org/abs/0808.2596>

²² Uniqueness Criteria for the Virasoro-Shapiro Amplitude

<https://arxiv.org/abs/2408.03362>

²³ <https://digital-strategy.ec.europa.eu/en/faqs/transparency-obligations-under-article-50-ai-act>

<https://digital-strategy.ec.europa.eu/en/faqs/transparency-obligations-under-article-50-ai-act>